CISS 143 Semester Project

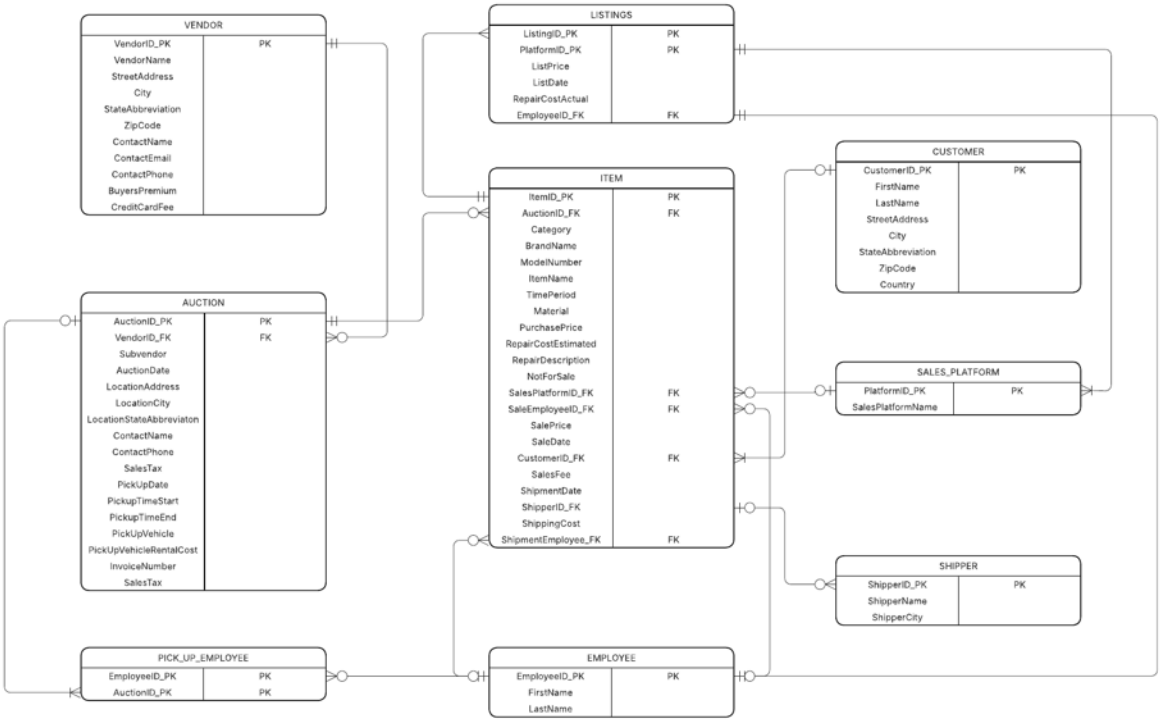
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Project Proposal

My semester project will be a database application designed for a small antique dealer utilizing Microsoft Access. The database will record data regarding inventory, employees, sales, refurbishment and restoration costs, marketing/sales costs, shipping costs and the procurement process by which the items are purchased at auction. The database will contain forms that simplify data entry regarding entities such as auctions, vendors, items/antiques, employees, and third-party sales platforms. The database will also generate reports regarding profits and losses, employee performance, and inventory status. These reports will process data into usable information that the user can turn into sound business decisions. The database will be used primarily by the antique dealer and its small number of employees as a replacement for the large and unruly spreadsheet that is currently being used. It will include a Dashboard as a main navigation point for the database. Additionally, it will feature an “autoexec macro” that will automatically launch the menu upon startup, allowing for easier data entry and report generation.

Database Structure

Entity Relationship Diagram



Relational Notation

ITEM (ItemID_PK, *AuctionID_FK*, Category, BrandName, ModelNumber, ItemName, TimePeriod, Material, PurchasePrice, AskingPrice, RepairCostEstimated, RepairCostActual, RepairDescription)

AUCTION (AuctionID_PK, *VendorID_FK*, Subvendor, AuctionDate, LocationAddress, LocationCity, LocationStateAbbreviation, ContactName, ContactPhone, PickUpDate, PickUpTimeStart, PickupTimeEnd, PickUpVehicle, PickUpVehicleRentalCost, InvoiceNumber, SalesTax, *PickUpEmployeEID_FK*)

VENDOR (VendorID_PK, VendorName, StreetAddress, City, StateAbbreviation, ZipCode, ContactName, ContactEmail, ContactPhone, BuyersPremium, CreditCardFee)

EMPLOYEE (EmployeeID_PK, FirstName, LastName)

SALES (SaleID_PK, ListDate, SoldDate, *PlatformID_FK, CustomerID_FK*, ListPrice, SoldPrice*, EmployeeID_FK, ShipmentID_FK*)

ITEM_SALE (ItemID_PK, SaleID_PK)

SHIPMENT (ShipmentID_PK, ShipmentDate, *SaleID_FK*, *ShipperID_FK,* ShippingCost)

SHIPMENT_EMPLOYEE (ShipmentID_PK, EmployeeID_PK)

SALES_PLATFORM (PlatformID_PK, SalesPlatformName, SalesFee)

CUSTOMER (CustomerID_PK, FirstName, LastName, StreetAddress, City, StateAbbreviation, Country)

SHIPPER (ShipperID_PK, ShipperName, ShipperCity)

Data Entry Forms

(1) Vendor Data Entry Form

Purpose: This data entry form will allow the user to enter data regarding a vendor/auction house.

Fields Involved:

VendorID (autogenerated)
VendorName
Subvendor
StreetAddress
City
StateAbbreviation
ZipCode
ContactName
ContactEmail
ContactPhone
BuyersPremium
CreditCardFee

Tables involved:

VENDOR

(2) Auction and Item(s) Won Data Entry Form

Purpose: This data entry form will allow the user to enter data in reference to a completed auction from which new inventory is won (purchased).

Fields Involved:

AuctionID (autogenerated)
VendorID
Subvendor
AuctionDate
InvoiceNumber
LocationAddress
LocationCity
LocationStateAbbreviation

PickUpDate
ContactName
ContactPhone
ItemID-autogenerated
Category (pick from dropdown list)
BrandName
ModelNumber
ItemName
TimePeriod (pick from dropdown list)
PurchasePrice
RepairCostEstimate
RepairDescription

Tables Involved:
AUCTION
ITEM
VENDOR

(3) Item(s) Picked Up Data Entry Form

Purpose: This data entry form will allow the user to enter data in reference to a specific item that is picked up from the auction site and added to the dealer's inventory.

Fields Involved:
AuctionID (pick from listbox)
EmployeeID (pick from listbox)

Tables Involved:
ITEM
AUCTION
VENDOR
PICK_UP_EMPLOYEE
EMPLOYEE

(4) Item(s) Listed for Sale Data Entry Form

Purpose: This data entry form will allow the user to enter data in reference to a specific item that is listed for sale via a third-party sales platform.

Fields Involved:

ItemID (pick from listbox)
ListDate
ListPrice
RepairCostActual
PlatformID
EmployeeID (pick from listbox)

Tables Involved:

ITEM
EMPLOYEE
SALES_PLATFORM

(5) Item(s) Sold Data Entry Form

Purpose: This data entry form will allow the user to enter data in reference to a specific item that is sold via a third-party platform. This form will contain a Subform that displays items in the inventory that are currently unsold.

Fields Involved:

ItemID (pick from listbox)
SoldDate
SoldPrice
CustomerID(autogenerated)
CustomerName
FirstName
LastName
StreetAddress
City
StateAbbreviation
ZipCode
Country

Tables Involved:

ITEM
ITEM_SALE
SALE
SALES_PLATFORM
CUSTOMER

(6) Item(s) Shipped Data Entry Form

Purpose: This data entry form will allow the user to enter data in reference to a specific item(s) that is shipped to a customer.

Fields Involved:

ItemID (pick from listbox)
ShipmentID(autogenerated)
ShipperID
ShipmentID
EmployeeID
City

StateAbbreviation
Country
ShippingCost

Tables Involved:

ITEM
EMPLOYEE
SHIPMENT
SHIPMENT_EMPLOYEE
CUSTOMER
SHIPPER

Reports

(1) Profit/Loss Report

Purpose: This report will generate information on profit/loss by year, by item category, by brand, by third-party sales platform, and by vendor/auction house.

Fields Involved:

VendorID
VendorName
Buyer'sPremium
ItemID
Category
BrandName
PurchasePrice
RepairCostActual
SaleID
SoldPrice
DateSold
ShippingCost
Fee
PickUpVehicleRentalCost

Tables Involved:

VENDOR
ITEM
ITEM_SALE
SALE
SALES_PLATFORM
SHIPMENT
AUCTION

(2) Vendor/Auction House Report

This report will generate information on the vendor/auction house and the inventory obtained from each of them.

Fields Involved:

VendorID
VendorName
StreetAddress
StateAbbreviation
ZipCode
ContactName
ContactEmail
ContactPhone
BuyersPremium
AuctionID
AuctionDate
LocationCity
LocationStateAbbreviation
ItemID
Category
BrandName
ModelNumber
TimePeriod
Material

Tables Involved:

VENDOR
AUCTION
ITEM

(3) Employees Report

This report will generate information on which employee has the most sales, and calculations regarding item pickups and deliveries.

Fields Involved:

EmployeeID
FirstName
LastName
SaleID
DateSold
SoldPrice
PurchasePrice
PickUpDate
PickUpVehicle

LocationCity

Tables Involved:

EMPLOYEE
ITEM
SALE
ITEM_SALE
AUCTION
PICKUP_EMPLOYEE

(4) Unsold Inventory Report

This report will generate information on unsold inventory.

Fields Involved:

ItemID
SaleID
DateSold
AuctionDate

Tables Involved:

ITEM
AUCTION
SALE
ITEM_SALE